

Optimizing Turnaround Time and Unlocking Off-Season Efficiency: A Dual Data-Driven Approach for Service Center Growth

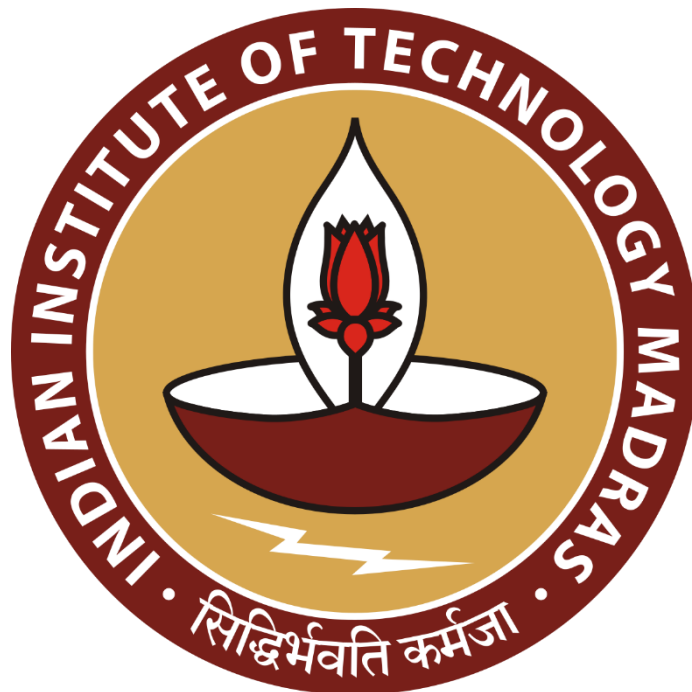
A Proposal report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project titled “Optimizing Turnaround Time and Unlocking Off-Season Efficiency: A Dual Data-Driven Approach for Service Center Growth”. I extend my appreciation to Prime Infotech, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

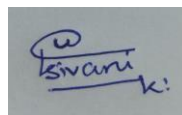
Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate:

A handwritten signature in blue ink, reading "Sivani Krishna T" with a stylized flourish above the name and a horizontal line below it.

Name: Sivani Krishna T

Date: 12/06/2025

1 Executive Summary and Title

Title: Optimizing Turnaround Time and Unlocking Off-Season Efficiency: A Dual Data-Driven Approach for Service Center Growth

This project examines Prime Infotech, an authorized computer service center located in Calicut, Kerala. The company primarily operates as a B2B service provider through brand partnerships providing warranty logistics and servicing support behalf of the brands, while also fulfilling a consumer-facing role by handling service logistics for end users .Committed to high service standards, Prime Infotech bridges the gap between brands and end-users by ensuring seamless, efficient support.

Two major problems faced by the company are delays in warranty claim turnaround time (TAT) and underutilization of resources during the summer season. Inconsistent TAT across different brands often leads to SLA breaches, delayed payments from brands, and reduced operational efficiency. At the same time, the company sees a noticeable dip in service volume during summer months, resulting in idle technician capacity and lower revenue flow. These seasonal inefficiencies, combined with inconsistent claim processing, affect overall profitability, brand satisfaction, and the service center's ability to operate at full potential throughout the year.

The issues will be addressed by analyzing operational and claim data to generate actionable insights. Descriptive analysis will identify average TAT per brand and flag SLA breaches. Seasonal trend analysis will highlight low-volume periods, especially in summer, to improve technician utilization. The goal is to streamline processes, reduce delays, and use downtime for preventive maintenance or training. These improvements aim to boost brand satisfaction, ensure consistent revenue, and enhance the overall profitability of the service center.

2 Organization Background

Prime Infotech was established in 2012 as a partnership between the current owners, Mr. Sajeev Kumar T, and Mr. Sibi K Thomas. It originally operated at Pottammal Junction, Calicut, Kerala, and functioned as more than just an authorized service center. In its early years, the company provided both onsite hardware and software support services, employing around 10 service personnel who covered the Calicut, Malappuram, Wayanad and Kannur regions.

Since 2022, Prime Infotech has been operating from its current location at Chinthavalappu, Palayam Junction, Calicut. Today, the company primarily functions as an authorized warranty

service provider for several leading technology brands. It operates as a neutral service center, not engaged in direct sales, and is now exploring business expansion opportunities particularly in projector rental and servicing to diversify its offerings and improve revenue streams

3 Problem Statement

3.1 Problem statement 1: Delays in warranty claim turnaround time (TAT) are leading to frequent SLA breaches and delayed brand payments, directly affecting service efficiency and cash flow

3.2 Problem statement 2: Seasonal drops in service volume during summer months result in underutilized technician capacity and idle inventory, lowering productivity and profitability.

3.3 Problem statement 3: Limited diversification of revenue streams and low marketing visibility make the service center overly dependent on brand-authorized claims, reducing resilience during low-demand periods and restricting overall business growth

4 Background of the Problem

The company's warranty service operations face critical challenges due to unpredictable and prolonged turnaround times (TAT) in claim processing across different partner brands.

Although the service center submits claims promptly, delays often arise at the brand's end such as slow approvals or replacement shipments which breach service-level agreements (SLAs) and delay payments. This affects customer satisfaction, increases complaint volumes, and disrupts cash flow

Strong seasonal variation in service volume is also an issue. During the summer months, the number of incoming warranty claims drops significantly, leading to underutilized technician, idle inventory, and reduced revenue. In contrast, the rainy season typically sees a surge in service requests, often resulting in resource strain. This imbalance creates planning challenges and prevents consistent business performance year-round.

A concern is the company's heavy dependence on brand-authorized claims as its primary revenue source. Since no direct payments are collected from customers, and brands do not always pay promptly, any delay or dip in claim volume directly impacts financial stability. Limited marketing efforts and lack of diversified income streams further restrict growth opportunities. These internal inefficiencies and external dependencies highlight the need for

data-driven insights and strategic business development to improve revenue and operational resilience

5 Problem Solving Approach

To find the possible reasons behind delays in turnaround time (TAT) and seasonal drops in service volume, we will carry out data-driven analysis using the company's historical service records. These two core issues impact both operational efficiency and revenue generation. By examining the patterns in claims handling and customer inflow across seasons, aiming to derive actionable insights to support decision-making and performance improvement.

Data Collection: Data will be collected for a period of 6 months from internal service center logs and Excel sheets set up by the business.

Methods of analysis:

- Descriptive Analytics: To calculate average TAT by brand, identify trends, and summarize seasonal claim patterns.
- Process Mapping: To visualize the full warranty claim lifecycle and highlight bottlenecks.
- Root Cause Analysis (RCA): To investigate major sources of delay (internal or external).
- Trend Analysis to identify seasonal surges in product inflow
- Time Series forecasting to plan for technician and inventory utilization
- Correlation mapping to study if external factors (like weather or festival seasons) impact inflow volume

Required tools:

The primary tool used for analysis will be: Microsoft Excel, as it allows for efficient data cleaning, calculation, and visualization through pivot tables, charts, and basic statistical functions. For more advanced insights and dynamic reporting, Power BI may be optionally used to create interactive dashboards that highlight trends, SLA breaches, and seasonal patterns. These tools are sufficient for analyzing turnaround time, forecasting claim volumes, and supporting data-driven decision-making without requiring programming expertise.

6 Expected Timeline

6.1 Work Breakdown Structure:

6.2 Gantt chart:

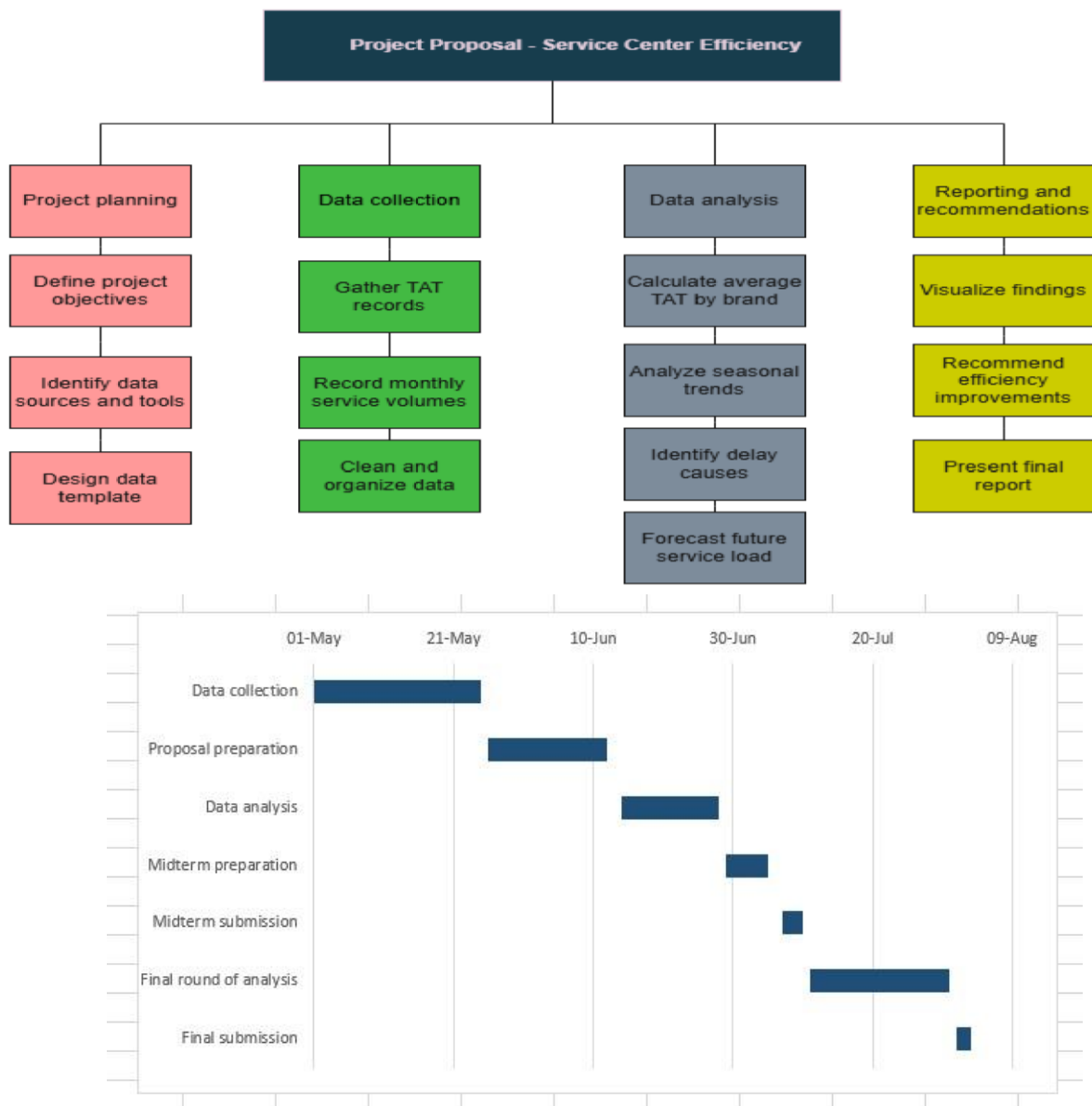


Figure
1

Expected timeline for completion of project

7 Expected Outcome

- Better TAT management

By analyzing the turnaround time (TAT) of warranty claims across different brands, the company will gain clear visibility into which brands or claim types contribute to the most delays. This will help in identifying weak points in the process such as shipping delays, slow brand response times, or internal handling issues enabling the service center to take corrective action and reduce SLA breaches. Better TAT management will improve cash flow, brand satisfaction, and operational reliability.

- Seasonal forecasting and resource optimization

Seasonal analysis will reveal predictable inflow patterns of product service requests, particularly the drop during summer and surge during monsoon. This insight will help in resource planning, such as adjusting technician shifts, managing spare part inventory, and reducing idle manpower or inventory.

- Revenue opportunities

Additionally, the study may open up opportunities to explore alternate revenue streams during lean months like marketing projector rental services, offering paid preventive maintenance, or partnering with new brands. Overall, the expected outcome is enhanced efficiency, improved SLA compliance, optimized resource utilization, and actionable strategies for boosting revenue even during off-peak service seasons.